Partha Chakraborty

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SUMMARY

Software engineer with 2+ years at startups building API integrations, workflow automation, and LLM features on Python, FastAPI, Node.js, and React, plus a deployed multi-agent RAG app. M.S. in AI and Machine Learning, University at Buffalo, expected January 2027.

EXPERIENCE

TurboMail **Buffalo, NY (Remote)**
AI Engineer Intern *May 2026 – Aug 2026*

- Wrote bulk email deletion and folder-move APIs in FastAPI, plus email-body sanitization and multi-account thread scoping.
- Automated unsubscribe handling with FastAPI background tasks that parse RFC 8058 List-Unsubscribe headers.
- Designed the database schema and APIs for subscription and payment tracking, then built the customer-facing React interface.
- Owned production features across the stack in React, TypeScript, Python, FastAPI, SQLAlchemy, and SQL.
- Fine-tuned Qwen2.5-7B with LoRA and PEFT on email tasks, raising classification F1 from 18.9% to 87.6% and ROUGE-L from 20.3 to 39.7 for summaries and 24.0 to 40.1 for replies; also beat Mistral-7B.

Bird Vision AB **Remote**
Software Developer *Jun 2023 – Jul 2025*

- Integrated Eboka with the company's other products end to end, so customers got value from each in one subscription cycle.
- Built two-way customer data sync between two company apps, with scheduled background bulk imports that notify users.
- Added on-demand single-customer import by user ID in both directions, so users didn't have to enter the same data twice.
- Took backend features to production, connecting third-party APIs to existing workflows with Node.js and Express.js.
- Ranked second on the team in code contributions within a year, shipping frontend and backend work across three products.
- Wrote Cypress tests covering the main user flows in the web app.

EDUCATION

University at Buffalo **New York, USA**
M.S. in Artificial Intelligence and Machine Learning *GPA 3.9/4.0 | Expected Jan 2027*
Coursework: Algorithms, Database Systems, Data-Intensive Computing, Machine Learning

Presidency University **Bengaluru, India**
B.Tech. in Computer Science and Engineering *GPA 8.14/10 | 2018 – 2022*
Coursework: Computer Networks, Data Structures, Operating Systems, Computer Architecture, Engineering Mathematics

SKILLS

Languages: Python, TypeScript, JavaScript, SQL, Java
Backend and integrations: FastAPI, Node.js, Express.js, REST APIs, SQLAlchemy, Redis, API integration
AI and LLMs: Claude API, Gemini API, agent orchestration, agent evaluation, RAG, tool calling, Sentence Transformers, embeddings, vector search, BM25, cross-encoder reranking, PyTorch, LoRA, PEFT
Data and tools: PostgreSQL, MySQL, MongoDB, Firebase, ChromaDB, Docker, Linux, Git, GitHub Actions, Pandas
Frontend and testing: HTML, CSS, React, Next.js, Redux, Tailwind CSS, React Native, Cypress, pytest
Development tools: Tree-sitter, Cursor, Claude Code, GitHub Copilot

PROJECTS

AI Codebase Intelligence [Live Demo] [GitHub] Python, Gemini, ChromaDB, Streamlit

- Deployed a Gemini multi-agent RAG app for GitHub and local repos: Architect, Code Reviewer, Bug Finder, Docs agents.
- Grounded answers in code with Tree-sitter chunking in six languages, hybrid semantic/BM25/symbol retrieval, and reranking.
- To keep answers reliable, built a source explorer with file paths, line ranges, and citation checks; reached 87.5% Hit@5 and 0.750 MRR on an initial 8-question retrieval evaluation.

Greenlight [Live Demo] [GitHub] Python, Claude, Gemini, Streamlit, GitHub Actions

- Built a Claude/Gemini agent with tool calling to diagnose failing tests, edit code, and independently verify fixes.
- Restricted tools to the repository, blocked secret files, scrubbed test-process credentials, and kept tests read-only by default; enforced step, token, and cost budgets with per-run traces and reviewable diffs.
- Shipped a CLI, Streamlit demo, and GitHub Action for human-reviewed fix PRs; developed the project with Claude Code.
- Built hidden-test and regression-checking evaluations: a Gemini run solved 10/10 synthetic tasks and 4/13 attempted real bugs, requiring passing project tests and no test-file edits.

CERTIFICATIONS

Hugging Face Agents Course, Unit 1: Fundamentals of Agents [Certificate] Oct 2026